



Internship in the Space Debris and Clean Space office

Atmospheric modelling for re-entry applications and vision based navigation for rendez-vous

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Contents

1	Introduction	5
	Introduction	5
2	Space Debris Office	7
	SpaceDebris	7
2.1	Introduction	7
2.2	ESA re-entry tools	7
2.3	NRLMSIS2.1 atmospheric model	8
2.4	DTM2020 atmospheric model	9
2.5	Atmospheric models implementation	9
2.6	Validation of model implementation	10
2.6.1	The GOCE re-entry	10
2.6.2	The Cluster re-entries	11
2.7	Reconstruction of reference trajectory from TLEs	12
2.8	EDR methodology for drag-signal extraction	13
2.9	Validation of the EDR methodology on the POPACS orbit	15
2.10	Extension of the methodology to elliptical orbits	17
3	Clean Space Office	20
	CleanSpace	20
3.1	Introduction	20
3.1.1	The RISE mission	21
3.2	Dataset generation of a rendez-vous scenario	21
3.2.1	SurRender	22
3.2.2	Camera model	22
3.2.3	Scene description	22
3.2.4	Multi-Layer Insulation (MLI) model	23
3.2.5	Drift analysis	24
3.3	Detection algorithm: main principles	25

3.3.1	Filtering strategy	25
3.3.2	RANSAC strategy	26
3.3.3	Concentric pair selection and fit refinement	27
3.4	Detection algorithm: tracking mode	28
3.5	Pose estimation algorithm	29
3.5.1	Ellipse-to-circle correspondence	29
3.5.2	Pose solving	29
3.6	Results	30
3.6.1	Pose estimation: baseline dataset	30
3.6.2	Pose estimation: bias study	31
3.6.3	Pose estimation: drift dataset	33
3.6.4	Pose estimation: lower resolution dataset	34
3.6.5	Pose estimation: rescaled GEO satellite CAD	35
3.6.6	Pose estimation: partially occluded scene dataset	39
3.7	Validation against physical images	40
4	Conclusion	43
	Conclusion	43
	Bibliography	44

3 Clean Space Office

3.1 Introduction

Vision-based navigation (VBN) has become a key enabling technology for autonomous rendez-vous, proximity operations, in-orbit servicing (IOS) and active debris removal missions. By exploiting imagery from onboard cameras, VBN systems can estimate the relative position and orientation of a target spacecraft without requiring dedicated navigation aids on the client vehicle (so called unprepared targets).

Developing a robust VBN system for space applications remains challenging. Space scenes are subject to highly variable illumination conditions, with strong shadows, specular reflections and large variations in target appearance throughout a rendez-vous trajectory. Navigation algorithms must also remain reliable in the presence of image noise, limited onboard computational resources, partial occlusions, and uncertainties in the relative dynamics of the rendez-vous scenario. Furthermore, the availability of representative image datasets is extremely limited. Real flight imagery of close-proximity operations is scarce, while ground-based facilities can only reproduce a subset of the environmental conditions encountered in orbit. As a result, synthetic image generation has become an essential tool for the development and validation of VBN algorithms, introducing the additional challenge of ensuring that the generated imagery is sufficiently representative of real operational conditions.

This chapter presents the work performed in the context of RISE, ESA’s in-orbit servicing demonstration mission, to assess the feasibility of a mono-camera VBN solution for proximity operations. The target feature selected for navigation is the client’s Launch Adapter Ring (LAR), a passive geometric feature available on the vast majority of GEO satellites[5]. The study therefore addresses not only the design of a pose estimation algorithm, but also the generation of representative synthetic datasets and the characterization of algorithm robustness under realistic operational conditions, including illumination variations, relative drift and partial occlusions.

The chapter is structured as follows: section 3.2 describes the methodology developed to generate synthetic imagery representative of the RISE rendez-vous scenario, including the SurRender rendering pipeline, the camera model, and the modeling of the client satellite’s multi-layer insulation (MLI), whose high specular reflectivity poses a particular challenge for image processing. The main principles implemented in the detection and pose estimation algorithm are then presented in section 3.3, followed by the tracking mode developed to improve robustness under partial occlusions. A comprehensive results and discussion section is provided in section 3.6: performance is first established on a baseline dataset spanning the expected sun-illumination domain, before examining the sensitivity of the algorithm to biases through a reversed dataset, and then assessing robustness under translational and attitude drift of the chaser with respect to the client, as well as partial occlusions of the field of view caused by the robotic arm. Finally, the algorithm is validated against a dataset of physical images acquired over a full rendez-vous trajectory, providing a bridge between the synthetic training domain and real-world sensors.

3.1.1 The RISE mission

The RISE mission is ESA’s flagship in-orbit servicing mission, set to launch in 2029. The objective of the RISE mission is to provide life-extension services to geostationary (GEO) satellites that have depleted their propellant reserves but still have operational payloads. Once RISE has fulfilled its life-extension mission with a client it will provide the necessary thrust to bring the client to geostationary graveyard orbit (GGO) to safely dispose the client before returning to GEO to service its next client.

The first part of the mission is a technology demonstration of its docking capabilities under ESA operations. This phase of the mission will last a few months, during which time RISE will demonstrate its capabilities to safely approach a client satellite, perform docking activities, and take over its AOCS. This will include demonstration of key technologies such as the autonomous relative navigation system and the mechanical docking mechanism being developed for this mission. After successfully completing this stage, commercial operations by the Prime contractor D-Orbit will follow, with the aim of demonstrating the commercial viability of life extension in orbit as a service.

In October 2024, it was announced that the Italian new-space company D-Orbit would be awarded the development contract. The RISE project is currently in Phase B2, having passed the System Requirement Review (SRR) in September 2025 and with the Preliminary Design Review (PDR) scheduled for the end of 2026.

3.2 Dataset generation of a rendez-vous scenario

Developing and validating a vision-based navigation algorithm for a mission such as RISE requires a large volume of representative imagery, covering the full range of operational conditions that may be encountered in flight. Acquiring such data from real in-orbit imagery is not feasible as in-flight images cannot be obtained during the development phase, while ground-based test campaigns are costly and remain limited in their ability to precisely replicate the space-environment.

Synthetic image generation, therefore, becomes the primary means of producing training and testing data, using physics-based rendering tools capable of simulating the optical behavior of the scene with high fidelity. This approach introduces the risk of a domain gap: any imperfections in the camera model, lighting representation, or surface material properties, can degrade the performance of an algorithm when it is transferred from the synthetic training domain to real operational imagery, which is why ground based facilities such as ESA’s GRALS laboratory are still widely used for hardware-in-the-loop validation before an algorithm can fly.

The remainder of this section details specific components of the synthetic data generation pipeline developed for this study: the SurRender rendering software and the definition of a camera model, the baseline scene configuration used to represent the RISE rendez-vous geometry, the modeling approach for a realistic client satellite and the addition of a drift analysis to the rendez-vous scenario.

3.2.1 SurRender

The tool chosen to render proximity operations scenes was SurRender by Airbus Defence & Space. SurRender is an image rendering application designed specifically for space-based applications, allowing physically representative rendering of satellite rendez-vous scenes.

SurRender operates in a client/server architecture, where scene elements including object meshes, materials, illumination sources and the camera model are defined through Python scripts and passed to a physically-based raytracing core running on the server side. The engine simulates the interaction of light with surfaces and optics to produce radiometrically accurate renders (irradiance maps) in Wm^{-2} . SurRender has been used for the development and validation of image-processing algorithms for prior ESA mission, including JUICE and Mars Sample Return (MSR).

3.2.2 Camera model

The SurRender camera model provides a physically-based representation of the imaging system used to generate synthetic space scenes. It combines optical, detector and radiometric parameters to accurately reproduce how the sensor responds to incoming light. A sensor definition script was built to provide the specifications of the AuricamD80 wide-angle camera, which is currently used as the baseline for RISE. This camera provides 2048x2048 arrays with $5.5\mu m$ pixel pitch and features a 1.25mm lens, 10mm focal length, a 56.6° symmetric field of view. Additional parameters in the camera definition include the quantum efficiency (QE) at each spectral band from visible to NIR wavelengths, the lens transmittance, dark current and read-out noise. These parameters were sourced from the AuricamD80 datasheet and compiled into a dedicated SurRender sensor definition file, ensuring that the synthetic imagery reflects the actual radiometric and noise characteristics of the flight camera.

One of the most important properties of the camera model is the point-spread function (PSF) which characterizes the diffraction blur introduced by the optical system. The PSF causes the energy from a single point source to spread to neighboring pixels and is defined in the model through a Gaussian blur. Because the PSF directly governs the sharpness of the edges in an image, it can have a strong influence on image-processing algorithm that are sensitive to edge accuracy.

3.2.3 Scene description

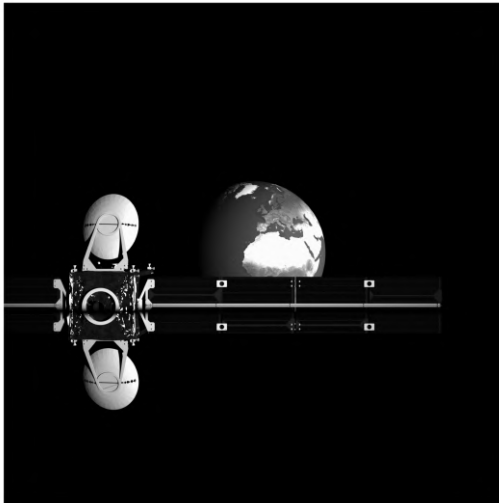
Two reference frames are used throughout this study: the world frame (based on SurRender's default world frame definition) and the RISE body frame with its origin at the geometric center of RISE. In the reference alignment, the two frames are coincident, with the camera's line of sight pointing along the negative world Z axis, the Y axis pointing toward the top of the image, and the X axis pointing toward the right of the image. Yaw, pitch, and roll describe the orientation of the body frame with respect to the world frame, defined respectively as rotations about the body Y, X, and Z axes. When drift is applied, the RISE body frame rotates relative to the fixed world frame according to these angles. The camera is rigidly fixed to the RISE

body, such that the camera frame shares the same orientation as the body frame at all times, with its origin offset to the camera's mounting location. Unless otherwise specified, all pose estimation results presented later in this chapter (translational distances and pitch/yaw angles) are expressed in this camera frame.

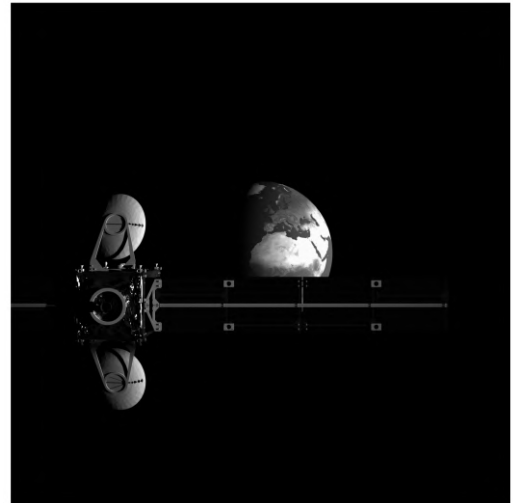
The baseline scene description for the RISE rendez-vous scenario flows down from the approach and docking concept and mission analysis. The scene parameters are provided below. The 2m client range corresponds to the standard capture distance during which the VBN algorithm will be operational to guide the capture sequence, while the sun azimuth and elevation ranges reflect the expected orbital geometry and illumination conditions during the full duration of the capture window.

- Sun azimuth $\alpha = 40$ to 80 degrees
- Sun elevation $\beta = 0$ to 10 degrees
- Range between the camera and LAR along the boresight = 2m
- Camera lateral offset from RISE body origin: [0.66m,0.23m]
- Client attitude is aligned with the RISE body frame

Figure 3.7 shows two representative examples from the resulting baseline dataset, illustrating the range of illumination conditions captured within this scene configuration.



(a) $\alpha = 45, \beta = 0$



(b) $\alpha = 75, \beta = 4$

Figure 3.1: Example of synthetic images from baseline with different illumination conditions within the range of expected mission conditions

3.2.4 Multi-Layer Insulation (MLI) model

The client satellite is to be modeled as a generic geo-stationary satellite. An important feature of this satellite is the presence of MLI on the structure and around the feature of interest,

the launch adapter ring. MLI has a high specular reflectivity coefficient, which can have a significant impact on image processing algorithms.

Blender was used to bake into the client satellite mesh a 3D map of an MLI texture, achieved by using a displace modifier. This 3D texture in combination with a highly reflective material definition within SurRender allows to model accurately the MLI by generating significant light scattering.

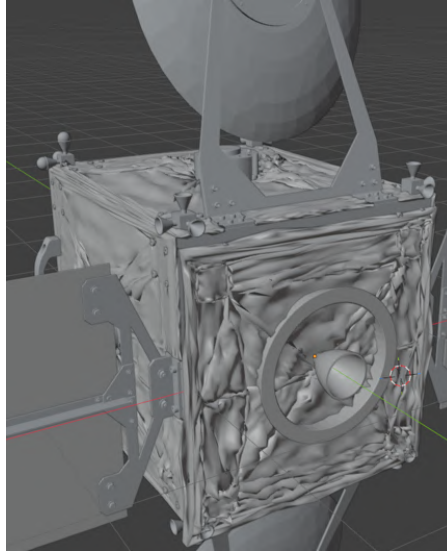


Figure 3.2: MLI texture

3.2.5 Drift analysis

During the capture sequence of the client satellite by RISE, the active attitude control will be temporarily deactivated. This option was selected as it simplifies the control of the robotic arm during capture which needs to be extremely precise in order to properly latch onto the client satellite's LAR. Indeed, disabling active control of RISE's attitude allows decoupling the control of the robotic arm and RISE, simplifying the software implementation greatly. On the other hand, this has an important consequence on the navigation system that must be studied. During the capture, any initial relative velocity between RISE and the client will cause a translational and attitude drift during the capture. This leads to the following constraints on the vision based navigation system:

- A maximum 3-axis attitude drift of 8° from the nominal capture attitude
- A maximum translational drift of 40cm in x and y

To analyse the algorithm's performance under these drift constraints, the full drift domain was studied by generating a dataset of 500 images, sampling combinations of random positions and attitudes within this domain. The strategy used was the Latin Hypercube Sample (LHS) which ensures the full range of parameters is studied. The dataset combines drift with the full range of positive and negative sun elevation and azimuth angles presented in subsection 3.2.3.

3.3 Detection algorithm: main principles

The client's launch adapter ring (LAR) consists of two concentric, coplanar circular edges, which project under perspective as two concentric ellipses in the image. Circular features have been widely studied for spacecraft pose estimation because their perspective projection remains an exact ellipse regardless of viewing orientation[6]. This geometric constraint is the central assumption exploited throughout the detection pipeline: edge fragments are filtered against the properties a genuine LAR arc must satisfy (subsection 3.3.1), candidate ellipses are fitted with a modified RANSAC scheme constrained by the same geometry (subsection 3.3.2), and the final pair is selected and refined by explicitly enforcing concentricity between the two fitted ellipses (subsection 3.3.3). This section provides a conceptual overview of the pipeline, describing the main processing steps and strategies employed along each stages of the pipeline.

3.3.1 Filtering strategy

The purpose of the filtering step is to reduce the raw image to a map of arc fragments, each belonging to a single conic, before any ellipse can be fitted. This stage of the pipeline uses a series of filters designed to exploit the geometric features of the LAR's two circular edges to discard any arc fragments that cannot belong to the feature of interest, such that the RANSAC stage described in subsection 3.3.2 can run on a small, information dense point pool.

Region of interest: Rather than searching the full frame, the edge map is first cropped to a symmetric window centered on the densest cluster of edge pixels, exploiting the specular reflectivity of the MLI material used on the client's surfaces. This step removes background edges including the Earth, solar arrays and high gain antennas while retaining the LAR.

Contour segmentation: Within the region of interest, contours are extracted from the edge-map and each one is split at an inflection point, where its local turning direction changes sign. A majority voting sliding window is used to distinguish a true inflection point from a noisy sign flip. This yields a pool of arc fragments that are each guaranteed to have the same convexity. Each arc is then inspected for abrupt changes in curvature radius. The metric used here is the three-point Menger radius, which measures the radius of the circle that passes through the three given points. An arc is again segmented at the location where the curvature radius change exceeds a threshold, resulting in a set of arcs that are convexity and curvature-consistent.

Curvature radius gating: The next step is to filter the remaining arcs based on their overall curvature. Near collinear arcs are first removed if their mean curvature is less than a threshold minimum curvature, removing large structural features such as solar or body panel edges.

Then, each remaining arc is evaluated against a curvature radius envelope. Here, the radius of curvature is no longer estimated from a local three-point sample as in the segmentation step above, but by fitting a circle to *all* points of the arc via least squares. The range of semi-major axis can be obtained from the requirements on the algorithm's depth operational range $[d_{min}, d_{max}]$ and the known radius of the LAR's inner and outer ring using Equation 3.1. The envelope of acceptable radii can then be derived from $[a_{min}, a_{max}]$ with some margin on either

side of the range.

$$a_{max} = \frac{f_x \cdot R_{outer}}{d_{min}} \quad a_{min} = \frac{f_x \cdot R_{inner}}{d_{max}} \quad (3.1)$$

At this final stage, a decimate function is applied to the remaining arcs if the number of points exceeds the target of 3000 points. The decimate function uniformly reduces the density of points in the arcs such that the number of points passed to the RANSAC stage is limited, allowing for predictability in the maximum computation effort at the RANSAC stage.

3.3.2 RANSAC strategy

The RANSAC algorithm forms the core of the ellipse detection pipeline. Rather than applying the standard RANSAC formulation, in which five-point samples are drawn uniformly at random from the entire edge-point pool, the sampling strategy was adapted to exploit the arc information produced by the preceding filtering stage. Specifically, each sample is drawn from a pair of arc segments rather than from the full point set. Since the filtering stage guarantees that each arc is both convex- and curvature-consistent, points sampled from one or two arcs have a much higher probability of belonging to the same physical ellipse than points selected uniformly from the complete edge map. This strategy is consistent with edge-linking and arc-grouping ellipse detection approaches reported in literature, where geometric relationships between arc segments are exploited to reduce the search space compared to classical methods such as the Hough Transform[7].

Sampling exclusively from a single arc was avoided because individual arcs typically span only a limited angular extent of the ellipse, resulting in poorly constrained fits that are more sensitive to noise. By instead sampling from two arcs, the spatial distribution of the sample is increased while maintaining a high likelihood that all points originate from the same ellipse. To further improve numerical stability, a minimum spatial spread is enforced, samples whose points are excessively clustered are rejected before ellipse fitting, preventing poorly conditioned models from progressing to the more computationally expensive inlier evaluation stage.

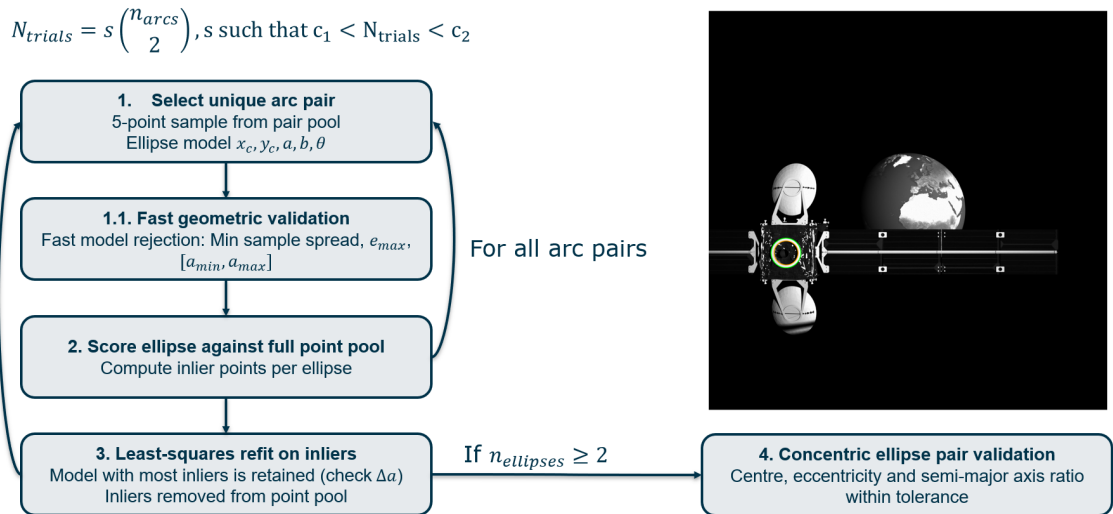


Figure 3.3: Simplified flowchart of the implemented RANSAC search algorithm

3.3.3 Concentric pair selection and fit refinement

The candidate ellipses retained after the RANSAC stage are then tested against four criteria to determine which pair of ellipses corresponds to the relative geometry of the inner and outer contours of the LAR. Given that the two circles are concentric, the center of their ellipse projections must coincide and they must have the same eccentricity. Additionally, the ratio of semi-major axis must fall close to the ratio of radii in the actual geometry. Since this ratio check requires both ellipses to be available simultaneously, both the inner and outer edges of the LAR must be detected within the same frame to reliably distinguish which candidate ellipse corresponds to the inner ring and which to the outer ring. Finally, a check is performed on the size of the retained outer ellipse against the algorithm's depth envelope as a final consistency check. Once an ellipse pair is retained, a final refinement is performed using the full edge-map (prior to filtering), tweaking the ellipse model to better fit boundaries that might have been over-filtered.

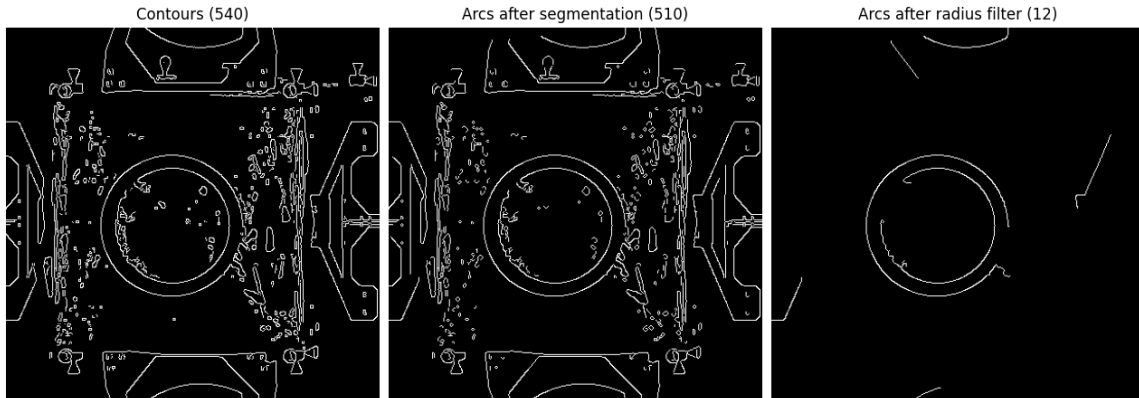


Figure 3.4: Edge map filtering stages including all contours in the search area (left), all arcs after convexity and curvature segmentation (middle) and the remaining arcs after collinear and radius filtering (right)

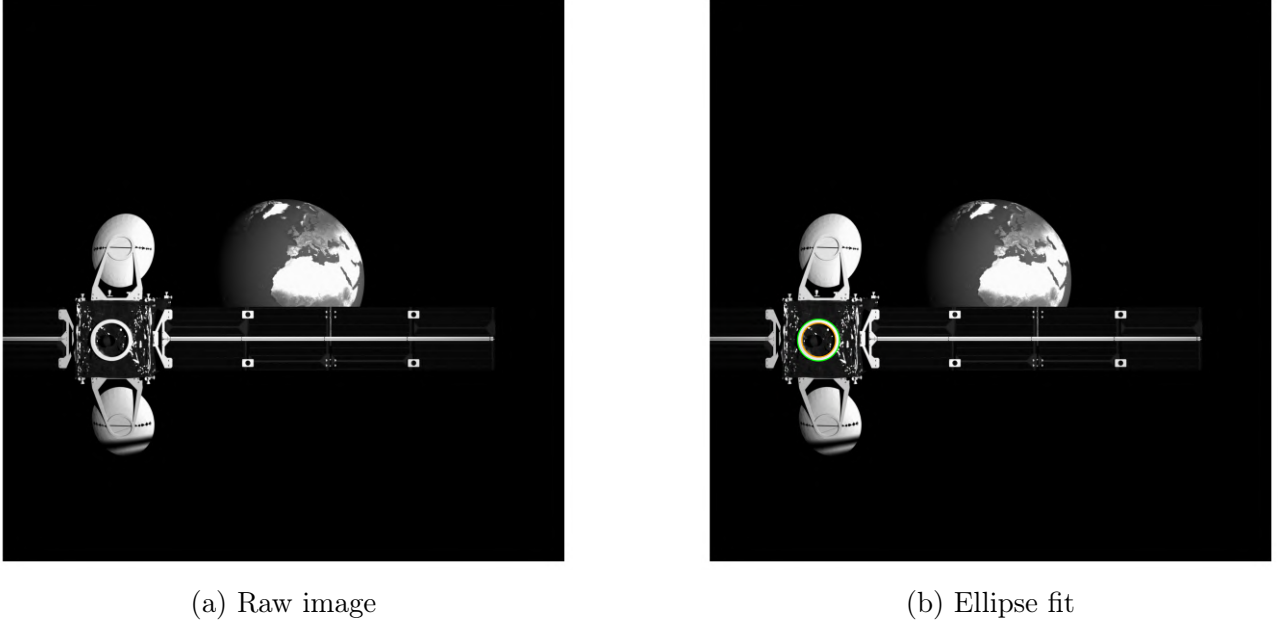


Figure 3.5: Algorithm result

3.4 Detection algorithm: tracking mode

During capture operations, the baseline scenario positions the LAR of the client satellite at a 2 meter distance from the WAC. During this phase, the WAC will acquire an image at 1Hz to feed the pose estimation algorithm. However, the operation of the robotic arm is expected to interfere with the field of view of the camera, leading to partially occluded views of the LAR. To deal with occluded scenes in a robust way, a tracking mode can be used where pose information obtained from previous images can be used to guide the detection algorithm on subsequent images.

Concretely, the tracking mode is integrated into the RANSAC-based detection algorithm by using the previous frame's ellipse estimate as a prior on the ellipse center location. Assuming a maximum chaser-client relative displacement of 5 cm over the 1-second interval between frames, candidate ellipse centers falling outside this bound are rejected before further processing. This requirement flows into the ellipse center search space constraint using the a priori depth and the camera's focal length:

$$r_{\text{px}} = \frac{f_x \cdot r_{\text{track}}}{d_0} \quad (3.2)$$

where r_{px} is the resulting radius, in pixels, of the search region within which the ellipse center is assumed to lie, f_x is the camera's focal length expressed in pixels, r_{track} is the maximum expected relative displacement between chaser and client over one tracking interval, and d_0 is the a priori depth (range) to the target. Any candidate ellipse center falling outside this bounding circle in the image plane is rejected prior to full RANSAC evaluation.

Given that obtaining the initial ellipse fit on the 5 point sample is much cheaper computationally than the evaluation of that ellipse against the full edge map, aggressively filtering the ellipse

before that step yields a substantial reduction in processing time, or for the same processing time a much larger number of potentially valid ellipse candidates can be evaluated.

3.5 Pose estimation algorithm

Once a concentric pair of ellipses has been detected and refined as described in subsection 3.3.3, the relative pose of the client with respect to the camera is recovered by solving a Perspective-n-Point (PnP) problem. Since the LAR’s inner and outer edges are known to be circles of fixed physical radius lying in a common plane, each detected ellipse can be related to its underlying 3D circle through a set of 2D-3D point correspondences, from which the camera pose can be recovered.

3.5.1 Ellipse-to-circle correspondence

For each detected ellipse, a set of 2D-3D correspondences is generated by sampling points uniformly along the fitted ellipse. Each sample point is translated to the ellipse center, rotated into the ellipse’s canonical frame using the fitted angle θ , and normalized by the semi-major and semi-minor axes a and b , mapping the ellipse onto a unit circle. This gives an angular parameter $\phi = \text{atan2}(y_{\text{norm}}, x_{\text{norm}})$ for each point, which is then used to place a corresponding 3D point on the known physical circle of radius R (the LAR’s inner or outer radius), lying in the $z = 0$ plane of a local object frame:

$$\mathbf{P}_{3D} = [R \cos \phi, R \sin \phi, 0]^T \quad (3.3)$$

This procedure is applied independently to the inlier points of both the inner and outer ellipses, using their respective known physical radii, yielding two independent sets of 2D-3D correspondences per frame.

3.5.2 Pose solving

The resulting correspondences are passed to OpenCV’s `solvePnPGeneric` function using the Infinitesimal Plane-based Pose Estimation (IPPE) algorithm, which is well suited to this problem given that all object points lie on a single plane. Because a planar point configuration can yield up to two geometrically valid pose solutions, symmetric about the plane, the physically consistent solution is selected by discarding any candidate that places the target behind the camera. The retained rotation matrix is subsequently converted to roll, pitch and yaw angles, following the axis convention defined in subsection 3.2.3.

Pose is estimated independently from the outer and inner ring correspondences, each producing its own translation vector, rotation matrix, and reprojection error, obtained by re-projecting the 3D object points back into the image using the estimated pose and comparing against the original inlier locations. In principle, this reprojection error could be used to select the more reliable of the two ring-based estimates on a frame-by-frame basis. In practice, however, the outer

ring consistently yields more accurate and stable pose estimates than the inner ring. The pose results presented in this chapter are therefore computed using the outer ring correspondences only.

3.6 Results

In this section, the performance results of the LAR detection and pose estimation algorithm will be evaluated on the various image datasets available. In subsection 3.6.1, the performance on the initial baseline dataset, where only the sun-illumination angle is varying, will be presented. In subsection 3.6.2, the algorithm is applied to an extended dataset including negative sun elevation and azimuth in order to study the behavior of the estimation bias. The performance is then evaluated on the dataset including translational and attitude drift in subsection 3.6.3. In subsection 3.6.5, the algorithm is again tested on the drift dataset but with a rescaled client satellite serving as a more representative client for the RISE mission. Finally, the performance on the dataset provided by D-Orbit including occlusions by the robotic arm and capture mechanism is presented in subsection 3.6.6.

3.6.1 Pose estimation: baseline dataset

The detection and pose estimation algorithm was run on the baseline dataset of 400 images covering the domain of sun azimuth and elevation (in 1° increments) provided by mission analysis at the 2m nominal capture range. Due to shadowing effects from the RISE's large solar wings obscuring the scene at low elevation ($\beta < 2^\circ$) and low contrast at large azimuth ($> 70^\circ$), it was decided to restrict the sun angles accordingly. In practice, the solution implemented for the mission to avoid these difficult lighting conditions will be to rotate the RISE solar wings by 90° in roll to avoid shadows on the target and the extreme sun azimuth will be avoided with the appropriate capture window.

In Figure 3.6 and Figure 3.7, the accuracy of the pose estimation model is provided with the sun-illumination domain restricted to 40 to 70 degrees in plane, 2 to 10 degrees out of plane. The observed estimation accuracy is excellent, and falls well within the needs of the robotics system for capture. A standard deviation $\sigma_z < 1mm$ is observed in range estimation, and in the order of $0.1mm$ in the xy plane was achieved on this dataset. In terms of angular pose, the estimation is within 1° of accuracy in both pitch and yaw. The roll estimation is not provided, as it is fundamentally unobservable from the LAR feature alone: the LAR consists of two concentric circular edges, and this geometry is rotationally symmetric about its perpendicular axis. A rotation of the client about this axis (roll) therefore produces no change in the LAR's projected appearance, regardless of the viewing geometry, making roll an inherently unobservable degree of freedom for a detection method based on the LAR's circular edges.

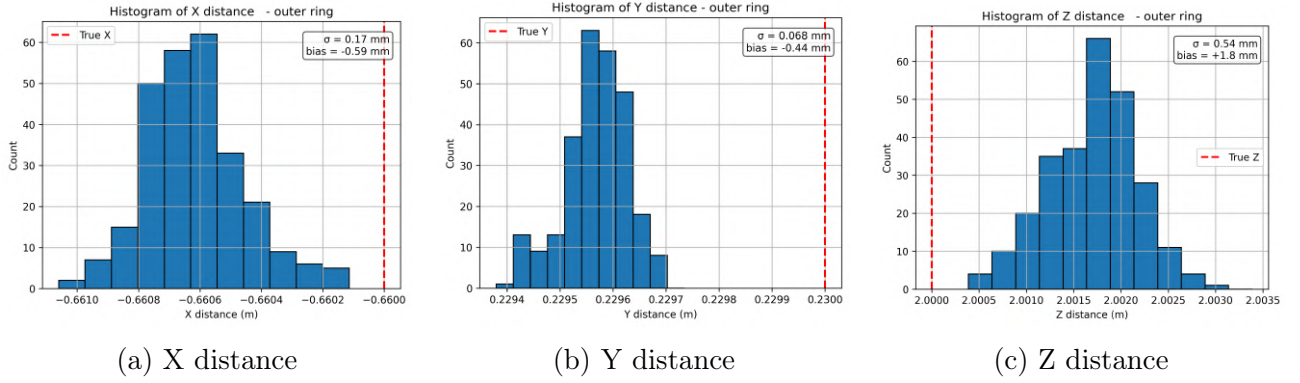


Figure 3.6: Translational pose estimation on baseline synthetic dataset

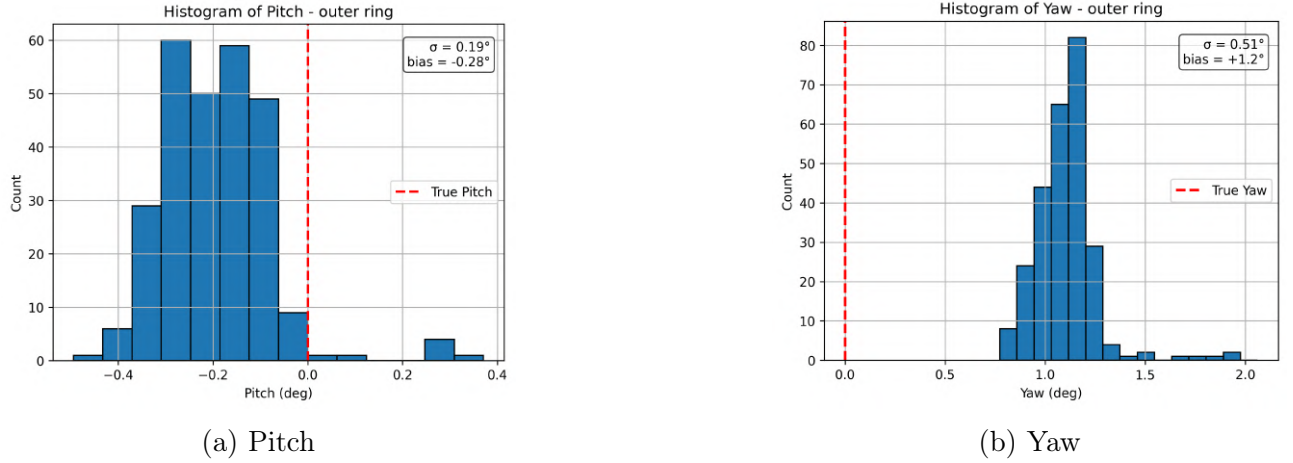


Figure 3.7: Angular estimation on baseline synthetic dataset

3.6.2 Pose estimation: bias study

The results on the baseline case demonstrate a significant bias (multiple times larger than the standard deviation) in the pose estimation results. It is interesting to understand what the bias is due to and whether a similar bias is observed across a range of conditions, in which case the bias could be corrected for within the attitude and position estimator. One of the limitations identified of the baseline dataset study case is the narrow range of illumination conditions, as only positive in plane and positive out of plane sun angles were considered.

The pose estimation results on the restricted illumination domain dataset, where high azimuth and low elevation angles were excluded as discussed in subsection 3.6.1, are provided in Figure 3.9 and Figure 3.10. As can be seen, the behaviour of the pose estimation bias is consistent between these two dataset, with a positive bias of around 1mm in range and biases in the x and y axis in the same direction across the two datasets.

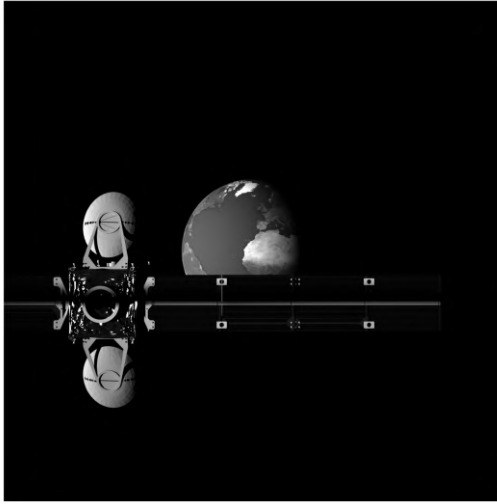
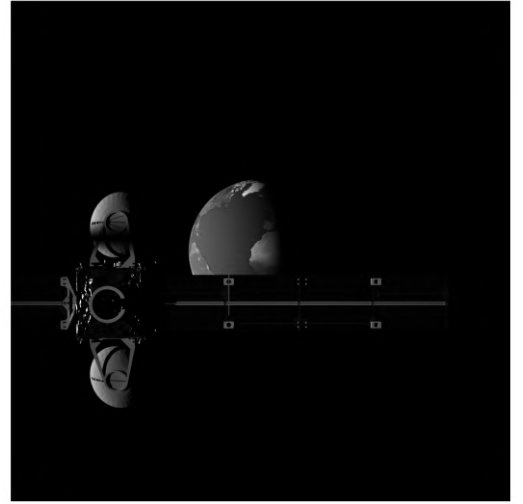
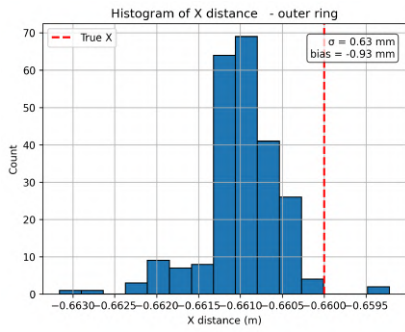
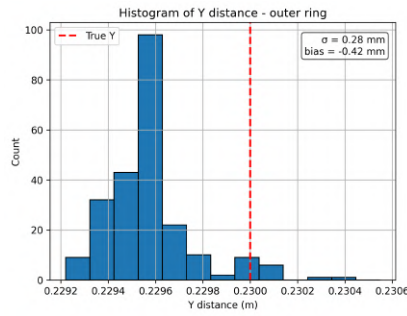
(a) $\alpha = -55, \beta = -1$ (b) $\alpha = -75, \beta = -8$

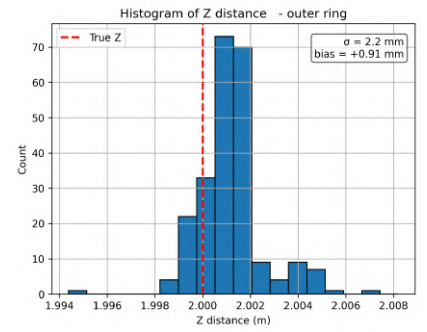
Figure 3.8: Example of synthetic images with reversed sun azimuth and elevation



(a) X distance

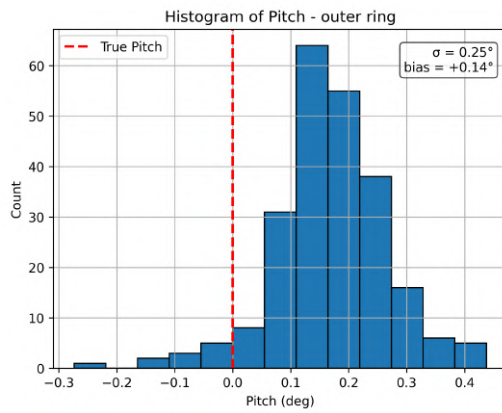


(b) Y distance

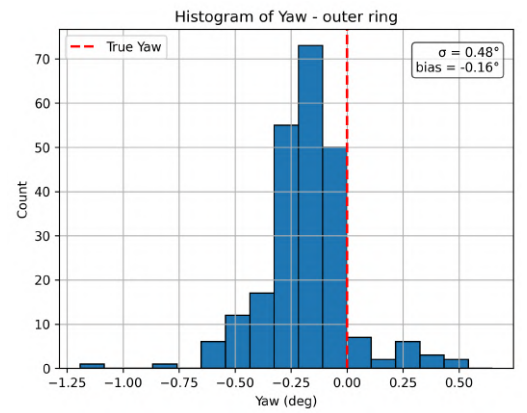


(c) Z distance

Figure 3.9: Translational pose estimation on synthetic dataset with reversed sun azimuth and elevation



(a) Pitch



(b) Yaw

Figure 3.10: Angular estimation on synthetic dataset with reversed sun elevation and azimuth

3.6.3 Pose estimation: drift dataset

To evaluate the robustness of the detection and pose estimation algorithm under realistic capture conditions, the drift dataset described in subsection 3.2.5 was used. This dataset contains 500 synthetic images generated using a Latin Hypercube Sampling strategy over the full translational and attitude drift domain, combined with positive and negative sun azimuth and elevation angles.

The resulting pose estimation accuracy is shown in Figure 3.11 and Figure 3.12. A detection rate of 99.4% is achieved across the dataset, demonstrating that the filtering and ellipse detection strategy remain robust throughout the operational domain. Translational accuracy remains excellent despite the increased variability of the scenes, with standard deviations below 1 mm in X and Y and 2.1 mm in range. Compared to the baseline dataset, the largest impact of drift is observed on the pitch and yaw estimates, whose error distributions exhibit a larger spread due to the increased variation in viewing geometry and perspective distortion. Nevertheless, the pose estimation remains sufficiently accurate across the full drift domain for the requirements of the robotics system, indicating that the expected translational and attitude excursions during capture are not a limiting factor for the proposed navigation approach.

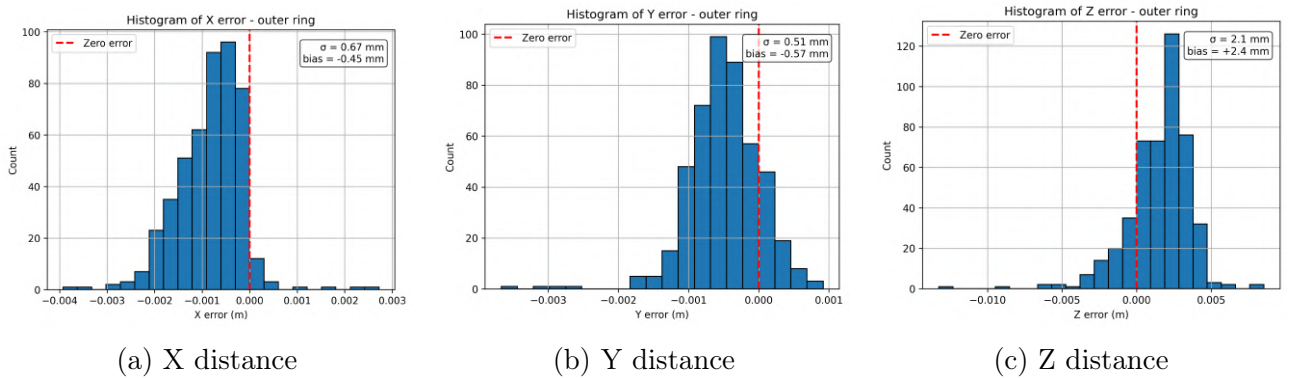


Figure 3.11: Translational pose estimation on synthetic dataset including translation and attitude drift

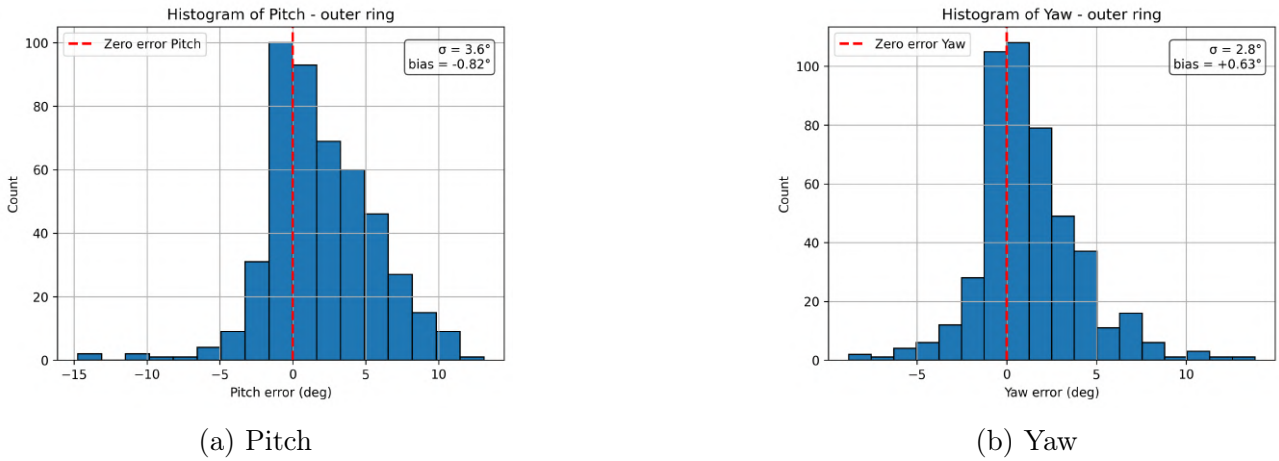


Figure 3.12: Angular estimation on synthetic dataset including translation and attitude drift

3.6.4 Pose estimation: lower resolution dataset

Binning is a common image processing technique for spacecraft cameras: adjacent pixels are combined to reduce the image resolution in exchange for lower data volume. Since bandwidth, data storage and processing power are tightly constrained, it is interesting to test whether the pose estimation algorithm remains robust when operating on binned imagery rather than full-resolution frames.

To evaluate this, 2×2 binning was applied to the drift dataset. Each output pixel is obtained by taking the average of a cluster of 4 pixels, reducing the 2048×2048 images to 1024×1024 . The LAR detection and pose estimation algorithm was run on this dataset to evaluate the sensitivity to this form of image degradation.

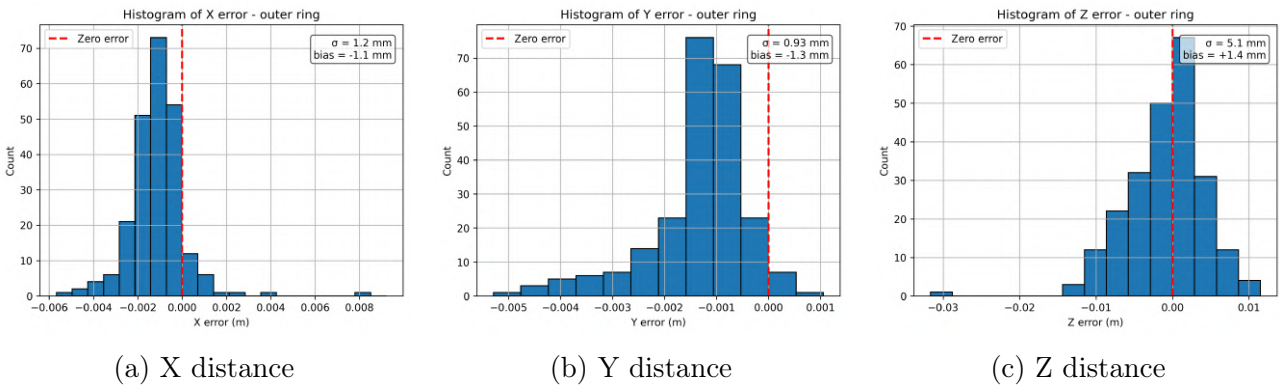


Figure 3.13: Translational pose estimation on synthetic dataset including translation and attitude drift

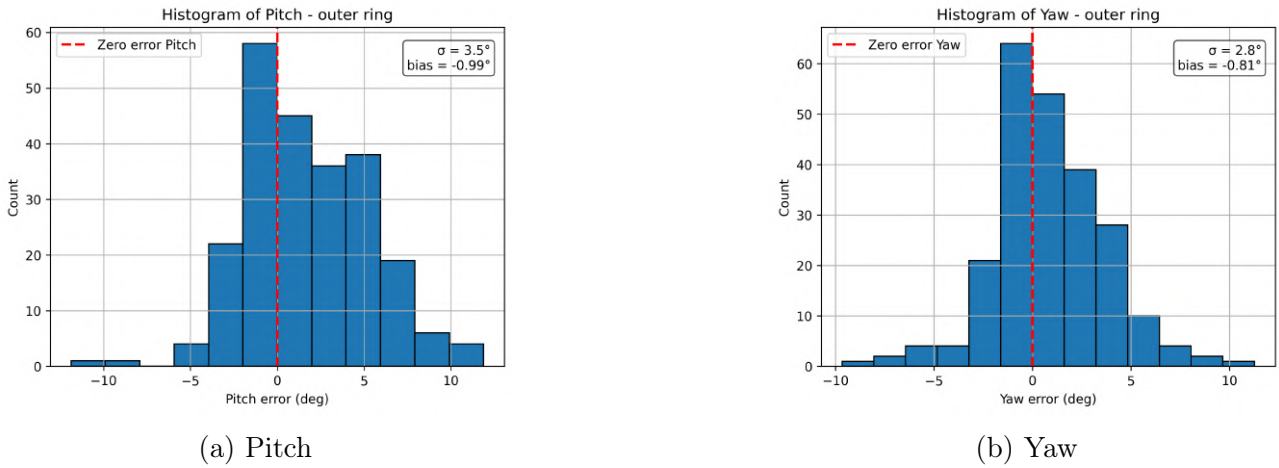
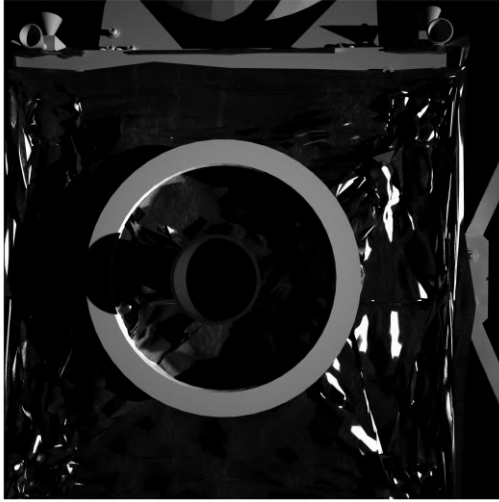


Figure 3.14: Angular estimation on synthetic dataset including translation and attitude drift

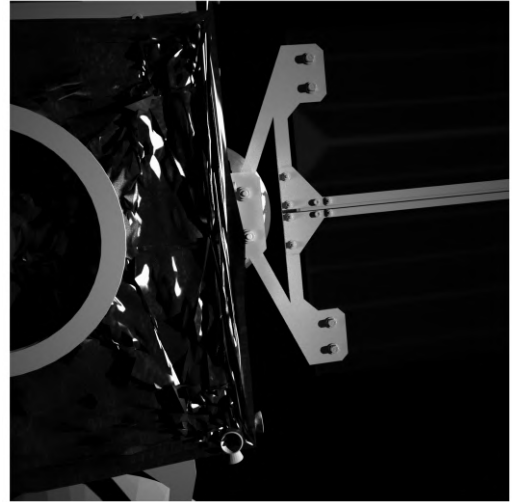
3.6.5 Pose estimation: rescaled GEO satellite CAD

The RISE mission will be operating in the GEO orbit, providing in-orbit servicing for large telecommunications satellites such as the Eurostar Neo or Spacebus Neo platforms. In order to test the algorithm in more representative conditions for the RISE mission, it was decided to generate a dataset of synthetic images with a larger client satellite comparable in size to the potential clients for RISE.

The CAD model of the generic GEO satellite used thus far was rescaled by a factor 7.4 such that its LAR matches the industry standard of 597mm used by most GEO telecommunication platforms. The same scene parameters used for the drift analysis dataset were used to generate the scenes in this dataset. Two examples taken from this dataset of 500 images are provided in Figure 3.15. In Figure 3.15b, the combination of translation and attitude drift parameters of that scene causes a portion of the LAR to exit the field of view at the standard 2m range. The influence of a partially invisible feature on the detection accuracy will be discussed in this section.

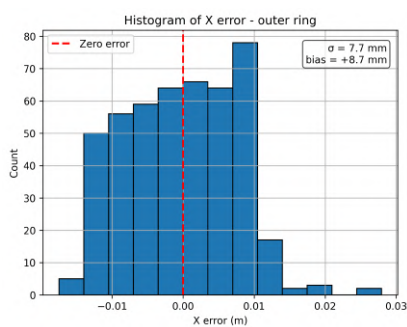


(a) Scene where the LAR is fully visible

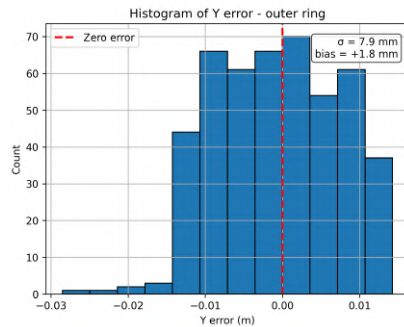


(b) Scene where the LAR is only partially visible due to a combination of translation and attitude drift

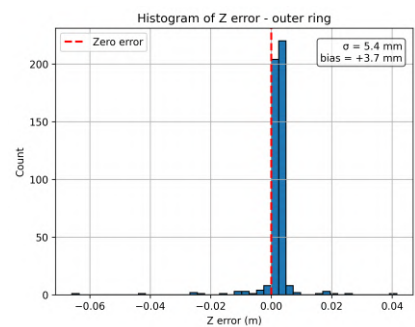
Figure 3.15: Synthetic images generated with the rescaled GEO satellite CAD



(a) X distance



(b) Y distance



(c) Z distance

Figure 3.16: Translational pose estimation on synthetic dataset with rescaled GEO client including translation and attitude drift

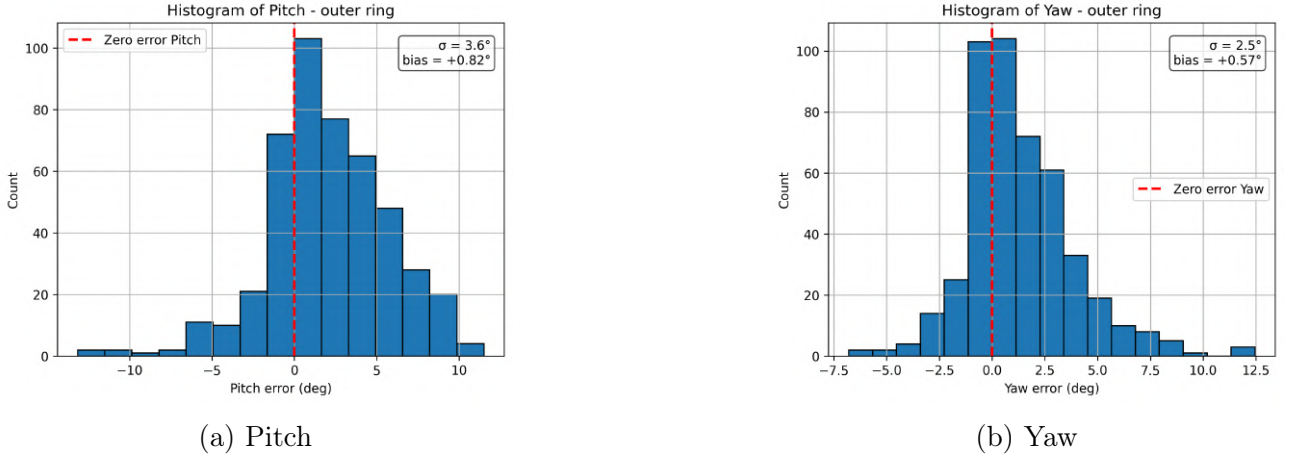


Figure 3.17: Angular pose estimation on synthetic dataset with rescaled GEO client including translation and attitude drift

The detection and pose estimation algorithm achieves 98.5% detection rate across the full dataset with the rescaled client, indicating that the larger LAR geometry does not significantly degrade the ability to detect this feature. In terms of pose accuracy, a slightly lower performance is obtained in the xy plane, with standard deviation of 7.7mm in X and 7.9mm in Y achieved. In range the results are still excellent with $\sigma_z = 5.4\text{mm}$. The range histogram has a much narrower shape than what was seen on other datasets, indicating that a few inliers in the range estimates are present with the error reaching up to 6cm on isolated scenes.

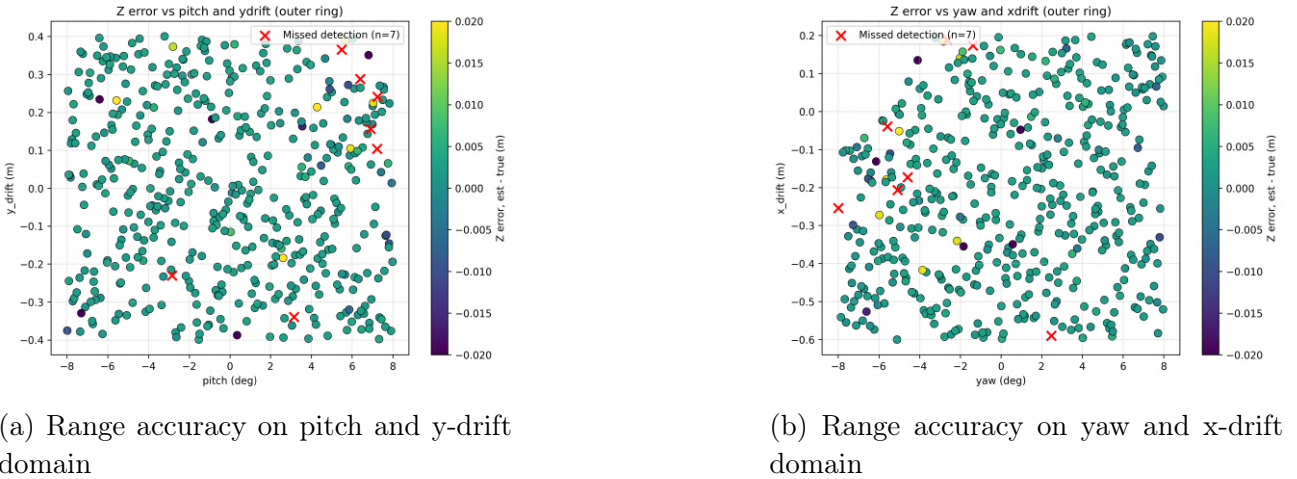


Figure 3.18: Range accuracy on drift domain. Significant accuracy drop near domain boundaries where the LAR is only partially visible

To investigate in further detail the detection accuracy exhibited by this dataset, a study of the drift domain was carried out, the results are shown in Figure 3.18. In these plots, the range error is shown as a function of the drift parameters, which are directly coupled: pitch and y_{drift} , and yaw and x_{drift} respectively. This study demonstrates that range estimates become more variable near the drift domain boundary when certain combinations of drift parameters cause

a significant portion of the LAR to exit the FOV. All seven missed detections observed in this dataset occur on frames where more than 40% of the LAR is out of the FOV.

An example of a lower visibility scene is presented in Figure 3.19, corresponding to scene 485 ($\text{yaw} = -7.2^\circ$ and $x_{\text{drift}} = -0.30m$), where only around 35% of the LAR outer ring is visible in the field of view. Under these conditions a Z error of -9mm is observed which is still an acceptable level of performance. On the visible portion of the ring, the detected ellipse fits the feature well, though the ellipse fit becomes under-constrained on the occluded side, where no edge information is available to anchor the fit. This illustrates a fundamental limitation of this approach with monocular vision under severe occlusion: beyond a certain loss of visible arc, the remaining edge information is insufficient to constrain the fit and the range estimate becomes less accurate.

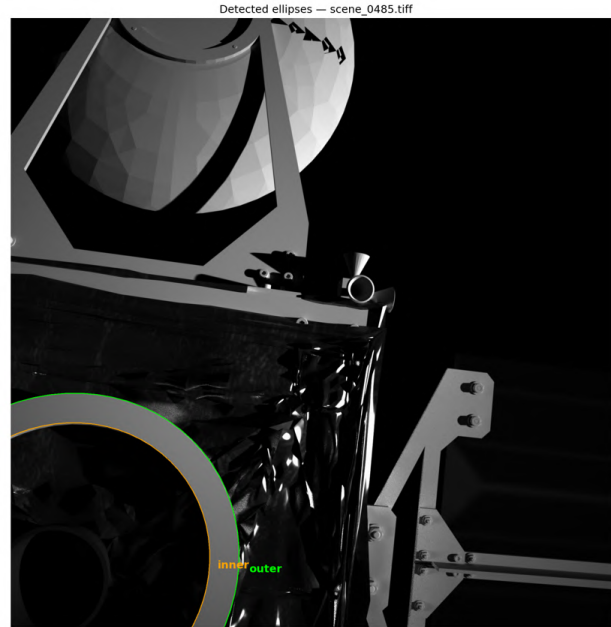
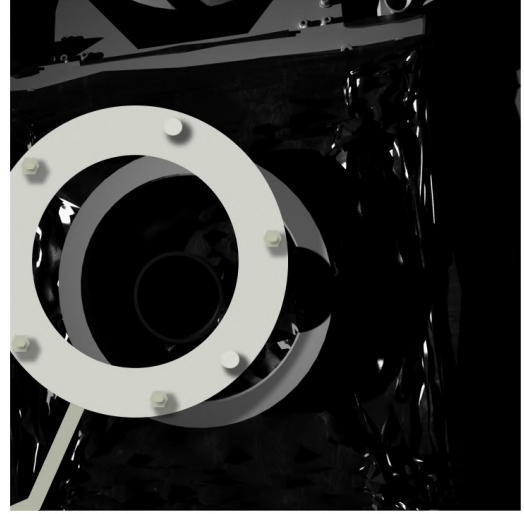


Figure 3.19: Detection example on a scene near the boundary of the drift domain, $Z_{\text{error}} = -9mm$

3.6.6 Pose estimation: partially occluded scene dataset



(a) Image with some occlusions from the capture ring during the capture sequence



(b) Image with large occlusions from the robotic arm and capture ring during the capture sequence

Figure 3.20: Synthetic images with a representation of occlusions of the LAR by the robotic arm and capture mechanism

During the capture sequence of the client satellite by RISE, the LAR may drift partially out of the FoV of the wide angle camera as discussed in the drift analysis in subsection 3.6.3. On top of this, the capture mechanism used by RISE to latch onto the LAR of the client will come into the FOV and may cause additional occlusions. D-Orbit provided a dataset of synthetic images where they simulated the capture sequence with the robotic arm and capture mechanism in frame. Unfortunately, these images are intellectual property of D-Orbit and thus cannot be shared in this report. A representation of a robotic arm and capture ring mechanism was added in post-processing to a synthetic image to demonstrate the extent of occlusions expected during the capture sequence, displayed in Figure 3.20a and Figure 3.20b representing different stages of the capture sequence.

As demonstrated in these images, the extent of occlusions is expected to increase throughout the capture sequence, with the full LAR exposed in the FOV at the beginning and occlusions appearing later due to a combination of the robotic interfaces and drifting. This is the typical scenario where a tracking mode is needed to ensure robustness of the detection and pose estimation throughout the sequence.

The results on this complex dataset are not yet at the required standard and further improvements to the detection algorithm will be needed to demonstrate robustness of the monocular vision solution under occlusions. Taking trajectory 5 and WCAM1 of the D-Orbit dataset for discussion, we obtain the following pose estimation results from the algorithm. A detection rate of 65% on the 20 image dataset, with the tracking mode being lost after scene 16 for the rest of the trajectory. By fitting a linear trajectory to the estimated pose, a standard deviation

of 21mm is obtained in range, 10mm in y and 5mm in x.

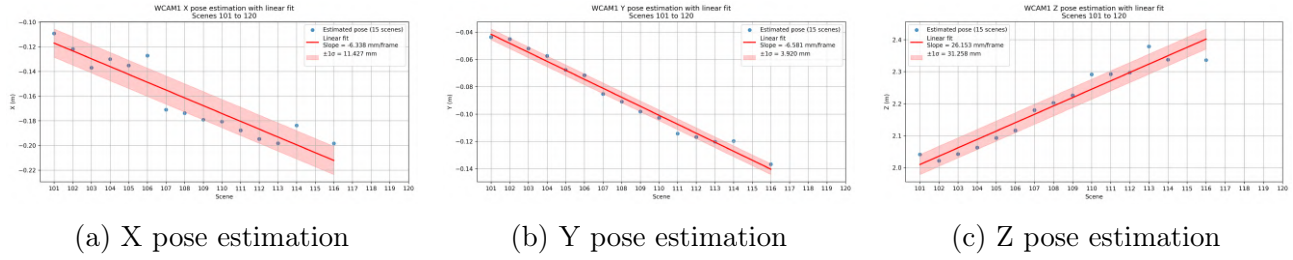


Figure 3.21: Pose estimation quality on the occluded dataset

3.7 Validation against physical images

The LAR feature detection and pose estimation algorithm was tested on a dataset of physical images acquired during a dedicated test campaign at ESA’s GRALS facility. The campaign reproduced a representative rendez-vous and proximity operations scenario, including a full approach trajectory toward a mock-up target. For the present analysis, only the close-proximity portion of the sequence was considered, corresponding to ranges between 1.6m to 1.0m. This subset contains 320 images acquired at a frequency of 1Hz.

Unlike the synthetic validation datasets presented previously, no ground truth pose information was available for this campaign because the facility testing was conducted by an external contractor. Furthermore, the acquired images are subject to confidentiality restrictions and therefore cannot be reproduced in this report. As a result, the assessment is limited to a qualitative analysis of the estimated trajectories and detection performance. Despite these limitations, validation on physical imagery remains particularly valuable because it can reveal discrepancies between reality and the assumptions embedded in the synthetic image generation process, including imperfections in the camera model, lighting conditions, material surface properties and the MLI model. As the dataset represents samples acquired along a continuous and physically smooth relative trajectory, it also provides an opportunity to evaluate the tracking mode developed in section 3.4 under realistic operating conditions.

Figure 3.22 and 3.24 present the estimated translational motion along the three camera-frame axes for the baseline and tracking-enabled algorithms, respectively. Visual inspection indicates that the estimated trajectories are temporally coherent and evolve smoothly throughout the sequence. On this dataset, no significant improvement in accuracy or detection rate is observed with tracking mode enabled. This can be explained mainly because the images from the lab don’t feature any background or additional features like antennas or solar arrays which are typically the features that can be eliminated with the tracking mode implemented in this algorithm.

Although no ground truth is available, the quality of the pose estimation can still be evaluated with respect to a low-order polynomial reference trajectory. Considering a relative motion characterized by approximately constant translational velocity and rotational rate between the client and the camera, quadratic trends naturally arise in the camera-frame coordinates. Con-

sequently, a second-order polynomial was fitted to the X and Y components. The Z coordinate, corresponding primarily to the range evolution during the approach, was found to vary approximately linearly and was therefore characterized using a first-order fit.

Although no ground truth is available to quantify the absolute accuracy, the residual dispersion around the fitted reference trajectories is very small. In the Z direction, the standard deviation of the residuals remains below 1 mm with respect to the linear reference trajectory. In the X and Y directions, the residual standard deviation is below 0.1 mm relative to the fitted quadratic trajectories. These results demonstrate that the algorithm is capable of producing stable and physically plausible pose estimates when applied to real imagery, despite the modelling inaccuracies and visual artifacts that are not captured by the synthetic image generation process.

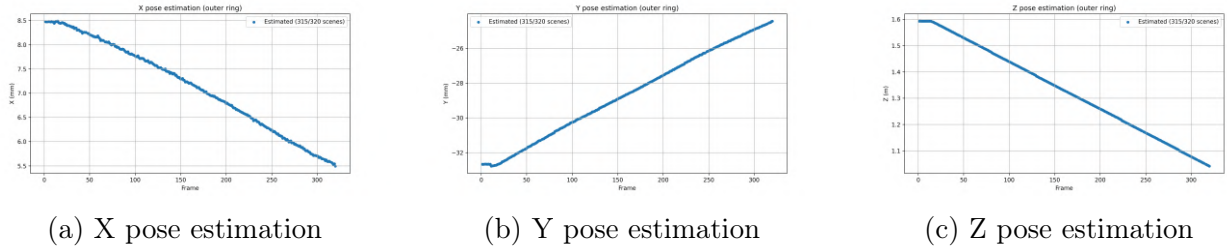


Figure 3.22: 3-axis pose estimation performance without tracking, detection rate 98.4%

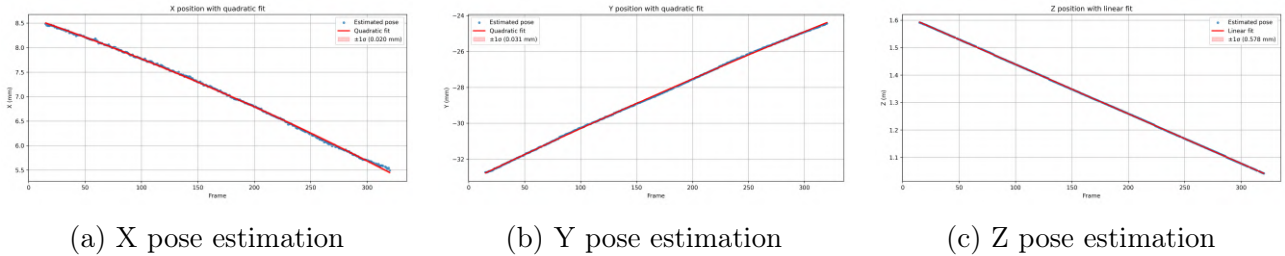


Figure 3.23: Pose estimation quality with respect to a quadratic fit reference trajectory

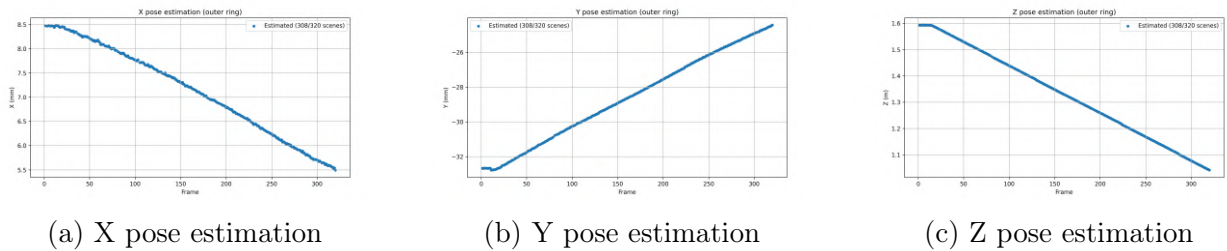
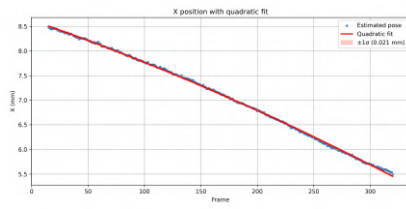
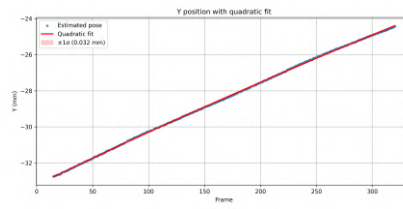


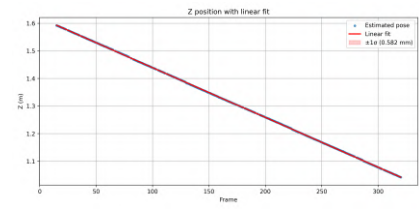
Figure 3.24: 3-axis pose estimation with tracking mode enabled, detection rate 96.3%



(a) X pose estimation



(b) Y pose estimation



(c) Z pose estimation

Figure 3.25: Tracking mode pose estimation quality with respect to a quadratic fit reference trajectory

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